Special right triangles

1 The side opposite the angle measuring 30°.

2
a $\frac{\sqrt{3}}{2}$
b $\frac{1}{2}$
c $\frac{1}{\sqrt{3}}$
d $\frac{1}{\sqrt{2}}$

3
a $\sqrt{3}$
b $x = 30^\circ$
c
i $\frac{\sqrt{3}}{2}$ ii $\frac{1}{2}$ iii $\sqrt{3}$
d
i $\frac{1}{2}$ ii $\frac{\sqrt{3}}{2}$ iii $\frac{1}{\sqrt{3}}$

Exact value trigonometric ratios

4
a 60°
b 0.5

5
a 45°
b 1

6
a 30°
b 0.5

7
a 60°
b 1.7

8
a 45°
b 0.71

9
a 30°
b 0.5

10
a 30°
b 0.9

11

a 30°

b 0.58

12

a 60°

b 1.73

13

a 60°

b 0.9

14

a 45°

b 60°

c 60°

d 30°

15

a $a=\sqrt{3}$

b $\theta=60^\circ$

16

a 45°

b 60°

17

a $x=\dfrac{\sqrt{30}}{2}$

b $h=1$

c $x=2$

d $P=17$

e $w=22$

f $x=4\sqrt{6},\ y=4\sqrt{3}$

g $u=8,\ v=4\sqrt{3}$

h $a=\dfrac{9\sqrt{3}}{2},\ b=\dfrac{9}{2}$

Applications

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a 60°

b 30°

c x units

d $x\sqrt{3}$ units

e i False. The hypotenuse of a 30°- 60°- 90° triangle is twice the length of the shorter leg.

ii True

iii False. The longer leg of a 30°- 60°- 90° triangle is $\sqrt{3}$ times the length of the shorter leg.

iv True

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a

i $p = \dfrac{5\sqrt{3}}{3}$ ii $q = \dfrac{10\sqrt{3}}{3}$ iii $r = \dfrac{5}{2}$ iv $s = \dfrac{5\sqrt{3}}{2}$

b $\dfrac{15\sqrt{3}}{2} + \dfrac{5}{2}$